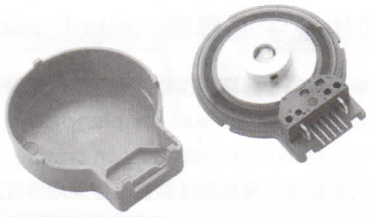
Rotary Encoder

HKT56xx Series



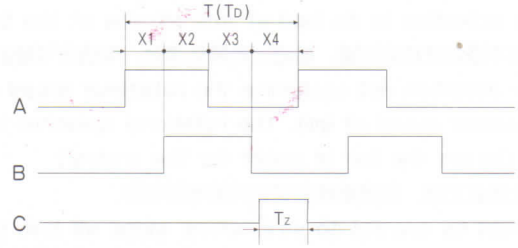
- 分体式设计，适应电动主轴、直流电机、伺服电机高转速、高频响、大批量使用的要求。
  Suitable for the minitype diret current electrical machinery,serve electrical machinery.
- 采用特殊矩阵数据读取，防尘、耐冲击、纠错能力强、可靠性高、工作寿命长。
  Strong ability load axle, it able to bear and assault anti-detonation well, Able to bear the high temperature, dependability is high,it is long.
- 普通型输出距离0.5m，加强型输出可达50m。
  The output distance 0.5m of the general type and the output distance 50m of the strengthen type.
- 提供定位模板、一体盘座光栅、模块、处理电路。
  Supply orientation module,raster,and module,disposing circuit.

技术参数 Technical Parameter

电源电压Vcc(Supply Voltage):	DC+5±5%
输出电压(Output Voltage):	高电平V_h≥85%Vcc,低电平V_L≤0.3V
消耗电流(Current requirements):	≤100mA
响应频率(Frequency response):	0~100KHz
输出波形(Output wave forms):	方波(Rectangle Wave)
载空比(Symmetry):	0.5T±0.1T
容许轴向游隙(windage of axis):	±0.1mm
容许径向游隙(windage of radial):	±0.015mm
转动惯量(Moment of inertia):	$5\times10^{-7}Kgm^2$
最大机械转速(Slew speed):	无限制(No have limit)
抗震动(Vibration resistance):	50m/s^2,10~200Hz,XYZ方向各2h(2hr each in XYZ directions)
抗冲击(Shock resistance):	980m/s^2,6ms,XYZ方向各2次(2times each in XYZ directions)
防护(Construction):	抗震、防尘 IP50(Equivalent to IP50 excl.receptacle)
工作寿命(The working life):	MTBF>20000h(+25℃,2000rpm)
工作温度(Operating temperature):	-10℃~80℃或-40℃~120℃
储存温度(Storge temperature):	-40℃~120℃
工作湿度(Operating humidity):	30~85% 无结露(unfrost)
重量(Weight):	0.02kg

输出波形与信号位置精度 Export the wave form and position recision of the signal

90° 相位差两信号与零位信号。从轴端看，顺时针旋转(CW)时的波形图。
Dual square wave in quadrature with index signal. This figure shous the waveforms when a shaft is rotated clockwise(CW) viewing encoder shaft.



波形比(Symmetry):X1+X2=1/2T±1/12T
X3+X4=1/2T±1/12T
相位差(Phase shift):Xn=1/4±1/12T(n=1、2、3、4)
信号位置误差(Division accuracy of signals):准确度(Accumulative angle error)≤1/18T
周期误差(Pitch error):±0.01T
零位信号(Signal width of Z channel):Tz=1/4T(1T、1/2T、1/4T……)
电角度周期(Electric angle period):TD=T
脉冲周期(Pulse period):T=360°/N(N为每转输出脉冲数N is the number of A、B channel pulses per revolution)
A、B相与Z相的位置关系不作规定(Positional relationship of A&B channels and Z channel is not specified)。

接线表(Connection Specifications):

排线代号No.	颜色标记1	2	3	4	5	6	7	8
导线颜色Cable color	黑Black	红Red	绿Green	棕Brown	灰Gray	白White	黄Yellow	橙Orange
L输出Output	OV	Vcc	SIG A	SIG $\overline{A}$	SIG $\overline{B}$	SIG B	SIG Z	SIG $\overline{Z}$
排线代号No.	颜色标记1	2	3	4	5	–	–	–
导线颜色Cable color	黑	黄	绿	红	白	–	–	–
E输出Output	OV	SIG Z	SIG A	Vcc	SIG B	–	–	–

注：实际接线以编码器的标签为准。
Note: Real connection should be obeyed to encoder's mark.

型号说明(The type proves)

- HKT | 56xx | W | — | 001 | G | 512 BZ 3 | / | 5 | E

HKT	56xx	W	001	G	512 BZ 3	5	E
产品型号 Model	外径轴径 outside dia axial dia	模块 Module	序列号 No.	航插·电缆 Plug·Cable	光栅规格数 Raster spec No.	电源电压 Voltage	输出方式 Output

- 型式举例 (An example of model):
  供货套件，壳体外径56，轴孔Φ8，模块W，电缆侧出，512p/r，DC5V电源，电压输出，Tz=$\frac{1}{4}$T。
  The rind radial Φ50, axial radial Φ8, module W, film material, side entry cable, 512p/r, supply voltage DC5V,the voltage output,Tz=$\frac{1}{4}$T.
  记为(Record for): HKT5608W-001G512BZ3/05E

HEDSS